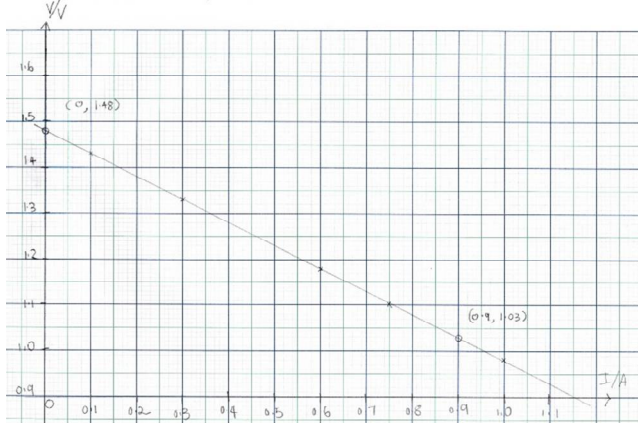


	$r = 2.41 \text{ } d \text{ (3sf)}$
5	
(a)(i)	current = $\frac{\text{total charge}}{\text{time}}$
	$= \frac{n \times A \times L \times e}{\text{time}}$
	$= nAve$
(a)(ii)	$I = nAve, v \propto \frac{1}{n(\text{diameter})^2}$
	$= 0.24 \text{ m s}^{-1} \text{ (2sf)}$
(b)(i)	
	
	Since, $V = E - Ir$, $V = -rI + E$ intercept = $E = 1.48 \text{ V}$
	gradient = $\frac{1.48 - 1.03}{0.00 - 0.90} = -0.50$
	$r = 0.50 \text{ } \Omega$

(b)(ii)	As temperature increases, resistance of thermistor decreases.
	The effective resistance of the circuit decreases. Current increases, causing terminal potential difference to decrease.
	Since $P = \frac{V^2}{R}$, power output of resistor R decreases.
6	